

SEDAR 21: HMS Dusky, Sandbar, and Blacknose Sharks

Dusky Shark

Data Workshop

LIFE HISTORY WORKING GROUP

- Increase research on post-release survivorship of all shark species by gear type
- Update age and growth and reproductive studies of dusky sharks.
- Develop empirically based estimates of natural mortality
- Continue tagging efforts

COMMERCIAL STATISTICS WORKING GROUP

No research recommendations were provided.

RECREATIONAL STATISTICS WORKING GROUP

No research recommendations were provided.

INDICES OF ABUNDANCE WORKING GROUP

No general research recommendations were provided. Recommendations specific to each index can be found in the workshop text and on the appropriate index scorecard.

CIE REVIEW RECOMMENDATIONS - DATA WORKSHOP

Conclusions and Recommendations

The Data Workshop provides a productive environment in which stakeholders and scientists can share knowledge to optimize the information available for assessment. It also serves as a mechanism where differences of opinion can be resolved before assessments are completed. The quality of science was high and appropriate for the purpose of stock assessment.

Compared with many stocks the availability of data are comparatively limited, especially in relation to catches, whether landings or discards. Although there is a large quantity of abundance index information the quality of these data is limited by the amount of fishery independent information or spatial coverage of the survey. Preliminary inspection of the indices at the meeting suggested that there was very little similarity of trends suggesting they have high uncertainty. There is a danger that the assessment might be driven arbitrarily by one of the time series if it happened to have low estimated CVs. I would **recommend that more exploratory analyses are done with the CPUE indices to try to identify those which contribute the most information on stock trends over the area of the assessment.** One possible line of analysis would be to use factor analysis to see if a common annual signal could be extracted from the suite of indices.

During the meeting some time was devoted to filling out a 'report card' for each series. In order to save time I **would recommend that the report card is completed by the author and that more time**

at the meeting is devoted to assessing the value of each time series for the assessment. The latter should include participation by assessment analysts.

The catch data suffer from a high degree of uncertainty. As much of the uncertainty relates to historical records there is not much that can be done to improve them. However, I would **recommend that an analysis is performed to try to quantify the uncertainty in the time series of catch data.** This would help in characterizing the overall uncertainty in the assessment.

The frequency of spawning by female sharks may be an important factor in estimating the spawning potential of the stock. Biological examination of female sharks appears to be able to determine that some species spawn less often than annually but the actual frequency cannot yet be established. In the absence of definitive information on spawning frequency I would **recommend that female sharks are examined in the spawning period to determine the proportion of spawning females.** While this will not provide an estimate of spawning frequency, it may provide sufficient information to estimate annual spawning biomass.

Estimates of discard survival proved an area of disagreement between scientists and fishing industry representatives. This was in part a result of differing perceptions of the meaning of discard survival. It is important that such disagreements don't lead to negotiated values that have no scientific basis. It might be worth investing in further discussion with the industry to reach a common understanding of the parameter in question. It might also help if **a desk study was undertaken to examine whether the choice of discard survival has a significant bearing on the estimated status of the stock in relation to MSY reference points.** If the sensitivity of the assessments to this quantity is low, it might defuse some of the polarization over the chosen values.

There may be a case for assessment analysts at the workshop to be more active in commenting whether certain biological effects can usefully be incorporated into assessments. This might be because some biological phenomena that are statically significant in their own right have little importance in determining the assessment outcome or where added biological realism in an assessment model is negated by the added uncertainty in input parameter values.

Assessment Workshop

The greatest source of uncertainty about dusky sharks is clearly the amount of human induced removals (e.g., discards) that are occurring. However, it is difficult to recommend a single course of action to improve this situation, as uncertainty in removals stems from a number of sources (species misidentification, non-reporting, etc.). Nevertheless, improving the reliability of removal data would help assessment modeling immensely.

Another suggestion for improving the reliability of assessment advice is the development of a stock-wide fishery independent monitoring program. The present assessment is based on a combination of spatially-restricted fishery independent surveys and several fishery dependent surveys. The former are not ideal in that observed trends may better represent localized dynamics than stock wide trends; the latter are deficient in that observed trends may often reflect changes in catchability (for instance, due to differences among vessels, captains, and changes in targeting) rather than absolute abundance.

Finally, further assessment work would benefit from a consistent life history sampling program that gathers annual samples of length and age-frequencies. The current hodgepodge of length-at-age samples is not sufficient to implement catch-age or catch-length models, and is only marginally useful for constructing selectivity curves because temporal changes in age-frequencies are confounded with

selectivity. Although an attempt was made to use existing age-length data to produce selectivity curves for the present assessment, this approach is clearly not ideal.

Review Workshop

The Assessment Team provided several research recommendations in the data workshop and AW reports, and these are endorsed by the Review Panel (RP) to the extent that they will improve the assessment. The RP considers research leading to an improved understanding of landings and removals, that improves consistency among indices, that reduces variability within the individual indices, and that leads to development or application of a model that more fully takes advantage of the length and age data including integration of the selectivity estimation into the assessment to be priorities.

With respect to further life history research, the RP considers the following to be priorities:

- Research on post-release survival by fishing sector and gear type should lead to improved landings and removals time series
- Research on fecundity and reproductive frequency should lead to an improved understanding of population productivity. As shown in assessment, status with respect to benchmarks is relatively robust to assumptions about overall productivity; however abundance and fishing mortality rate estimates are sensitive to this information. Research about natural mortality would also lead to a better understanding of productivity but traditionally has been difficult for most species.
- As noted throughout this report, the lack of age data was a limiting factor in this assessment and collection of sex-specific age and length data would aid the assessment. Regular collection of age data will help in the construction of improved age-length keys, in the interpretation of indices particularly in cases where populations have spatially structured with respect to age, and significantly aid in fitting the selectivity within the models. Additionally, if the abundance indices are age-structured, population responses to management actions should be detectable earlier than if the indices only provide information on total abundance.
- Although information about stock structure is important, as noted under ToR 1, genetic studies may not necessarily be informative about structure. Tagging studies to determine stock structure need to take into account that populations may be discrete during reproduction, but otherwise mixed most of the time. Increased international collaboration (e.g. Mexico) could help ensure wider distribution and returns of tags.

With respect to the abundance indices, the RP recommends:

- Evaluation of the individual indices via power analyses to determine whether they are informative about abundance trends. The majority of indices used in these assessments exhibited greater inter-annual variability than would be expected given the life history of these species, and given this variability, may only be able to detect large changes in abundance which are not expected to occur rapidly. A power analysis would help to determine how much abundance would have to change in order for the change to be detected with the survey, and additionally, if the survey effort needs to be

increased or re-distributed in order to be able to evaluate the effectiveness of rebuilding strategies given the relatively low population growth rates for these species.

- A small study on how to make the best use of the knowledge of the data workshop participants for developing index rankings.
- Ensuring that, to the extent possible, information about sex, length and age is collected for the reasons provided above.

With respect to the landings and removals, the RP recommends:

- Research that improves the understanding of historical landings, both in the modern and historical period and to support the assumptions about when stocks are at virgin biomass if this assumption is carried forward in future assessments. This is particularly important for GoM blacknose sharks given the difficulties reconciling the abundance indices, landings and life history information.
- As recommended by the AT, improved observer coverage particularly during periods of regulatory or gear changes (e.g. TEDs).
- Ensuring that, to the extent possible, information about sex, length and age is collected for each fishery in order that selectivity can be estimated in the model.

With respect to the assessment models, the RP recommends further model development using both simpler and more complex models taking the following into consideration:

- The RP noted that the models used in this assessment were reasonably suited to shark life history. However, other models (e.g. SS3) could also be adapted. If reproduction is modeled as a function of the number of mature females, uncertainty in the reproductive frequency, fecundity and pup-survival can be integrated into a single parameter (the slope at the origin of the SR function), and information about these traits can be incorporated via priors on the parameter. The RP recommends consideration of this approach if information on reproduction remains uncertain.
- Estimating the fishery and survey selectivities within the assessment model.
- Development of a two sex model for more direct estimation of the spawning stock
- Fitting the model to either length or age data. In addition to being necessary in order to estimate selectivities, these data can be informative about changes in age-specific abundance.
- Exploration of models that do not require an assumption that the population is at virgin levels at some point in time.
- If external age-length keys are used in future assessments, development of a key based on a growth model to better assign proportions-at-age in each length class.
- Simulation tests (management strategy evaluation) can be used to test the performance of alternative assessment methods (including the catch-free model, ASPM, ASPIC, SS3, or stock specific models), recruitment parameterizations, harvest control rules, assessment frequency and data collection. Simulation studies may have a particular use in these assessments because of the particular biology of sharks and the data poor nature of these stocks.

In the case of GoM blacknose shark, the appropriate interval of the next assessment depends on progress made towards reconciling the issues raised during this assessment process. For Atlantic blacknose shark, dusky shark and sandbar shark, the RP recognizes that population growth is expected to be relatively slow, but that modifications to the model may result in a different assessment of status. Benchmark assessments are recommended once the modifications are made. Additionally, for dusky shark, given the retrospective patterns in the present analysis and the resulting uncertainties in the assessment, updates using the existing model in the shorter term are also recommended. In the longer-term, development of a set of indicators (age-structure, total mortality estimates from catch curves, changes in abundance indices values) that could be used to determine whether status has changed sufficiently to warrant a full assessment, is recommended.

Sandbar Shark

Data Workshop

LIFE HISTORY WORKING GROUP

- Increase research on post-release survivorship of all shark species by gear type
- Continue to investigate reproductive periodicity for sandbar sharks
- Continue to collect vertebral samples from the sandbar shark research fishery to develop an ageing material archive and to keep track of the age distribution of the catch, and continue monitoring juvenile sandbar shark ages through the collection of fishery-independent samples
- Develop empirically based estimates of natural mortality
- Continue tagging efforts

COMMERCIAL STATISTICS WORKING GROUP

- Expand observer coverage to obtain 5% coverage of total trips or 20 to 30% PSE (percent standard error).
- Conduct more studies to better estimate post-release mortality
- Review bycatch estimation models
- Discard rates of sandbar sharks in the current directed and non-directed bottom longline fishery should be calculated and extrapolated using BLLOP data.
- Continue to develop better methods to quantify discards and effort from logbook programs and observer programs

RECREATIONAL STATISTICS WORKING GROUP

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Assessment Workshop

- Investigate alternative approaches to age-length keys for estimating age from length

Review Workshop

The Assessment Team provided several research recommendations in the data workshop and AW reports, and these are endorsed by the Review Panel (RP) to the extent that they will improve the assessment. The RP considers research leading to an improved understanding of landings and removals, that improves consistency among indices, that reduces variability within the individual indices, and that leads to development or application of a model that more fully takes advantage of the length and age data including integration of the selectivity estimation into the assessment to be priorities.

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- As noted throughout this report, the lack of age data was a limiting factor in this assessment and collection of sex-specific age and length data would aid the assessment. Regular collection of age data will help in the construction of improved age-length keys, in the interpretation of indices particularly in cases where populations have spatially structured with respect to age, and significantly aid in fitting the selectivity within the models. Additionally, if the abundance indices are age-structured, population responses to management actions should be detectable earlier than if the indices only provide information on total abundance.

- Although information about stock structure is important, as noted under ToR 1, genetic studies may not necessarily be informative about structure. Tagging studies to determine stock structure need to take into account that populations may be discrete during reproduction, but otherwise mixed most of the time. Increased international collaboration (e.g. Mexico) could help ensure wider distribution and returns of tags.

With respect to the abundance indices, the RP recommends:

- Evaluation of the individual indices via power analyses to determine whether they are informative about abundance trends. The majority of indices used in these assessments exhibited greater inter-annual variability than would be expected given the life history of these species, and given this variability, may only be able to detect large changes in abundance which are not expected to occur rapidly. A power analysis would help to determine how much abundance would have to change in order for the change to be detected with the survey, and additionally, if the survey effort needs to be increased or re-distributed in order to be able to evaluate the effectiveness of rebuilding strategies given the relatively low population growth rates for these species.
- A small study on how to make the best use of the knowledge of the data workshop participants for developing index rankings.
- Ensuring that, to the extent possible, information about sex, length and age is collected for the reasons provided above.

With respect to the landings and removals, the RP recommends:

- Research that improves the understanding of historical landings, both in the modern and historical period and to support the assumptions about when stocks are at virgin biomass if this assumption is carried forward in future assessments. This is particularly important for GoM blacknose sharks given the difficulties reconciling the abundance indices, landings and life history information.
- As recommended by the AT, improved observer coverage particularly during periods of regulatory or gear changes (e.g. TEDs).
- Ensuring that, to the extent possible, information about sex, length and age is collected for each fishery in order that selectivity can be estimated in the model.

With respect to the assessment models, the RP recommends further model development using both simpler and more complex models taking the following into consideration:

- The RP noted that the models used in this assessment were reasonably suited to shark life history. However, other models (e.g. SS3) could also be adapted. If reproduction is modeled as a function of the number of mature females, uncertainty in the reproductive frequency, fecundity and pup-survival can be integrated into a single parameter (the slope at the origin of the SR function), and information about these traits can be incorporated via priors on the parameter. The RP recommends consideration of this approach if information on reproduction remains uncertain.
- Estimating the fishery and survey selectivities within the assessment model.

- Development of a two sex model for more direct estimation of the spawning stock
- Fitting the model to either length or age data. In addition to being necessary in order to estimate selectivities, these data can be informative about changes in age-specific abundance.
- Exploration of models that do not require an assumption that the population is at virgin levels at some point in time.
- If external age-length keys are used in future assessments, development of a key based on a growth model to better assign proportions-at-age in each length class.
- Simulation tests (management strategy evaluation) can be used to test the performance of alternative assessment methods (including the catch-free model, ASPM, ASPIC, SS3, or stock specific models), recruitment parameterizations, harvest control rules, assessment frequency and data collection. Simulation studies may have a particular use in these assessments because of the particular biology of sharks and the data poor nature of these stocks.

Atlantic Blacknose Shark

Data Workshop

LIFE HISTORY WORKING GROUP

- Increase research on post-release survivorship of all shark species by gear type
- Update age and growth and reproductive studies of blacknose sharks, with emphasis on smaller individuals in the Atlantic and larger individuals in the Gulf of Mexico. Additionally, more information on litter size and reproductive periodicity is needed for blacknose sharks.
- Population level genetic studies on blacknose for stock discrimination(s).
- Develop empirically based estimates of natural mortality
- Continue tagging efforts

COMMERCIAL STATISTICS WORKING GROUP

The current level of shrimp trawl observer coverage is inadequate to model shrimp bycatch and catch rates with reasonable levels of uncertainty. The bycatch in the shrimp fishery also contains protected species and species of concern. With the current level of coverage, it is very difficult to statistically estimate bycatch of those rare species. More coverage would allow for better estimates of rare species, both protected and otherwise. We recommend the expansion of the observer program towards a goal of 2 to 5 % of the total effort. The recommended coverage levels are common in other observer programs, and have proved adequate for multiple types of statistical analysis. We recommend the program strive for even spatial coverage (particularly adding more south Atlantic coverage), randomness in vessel selection, and full identification to species of elasmobranchs (continuing on from the 2009 Bycatch Characterization Protocol).

RECREATIONAL STATISTICS WORKING GROUP

No specific research recommendations were provided.

INDICES OF ABUNDANCE WORKING GROUP

Specific research recommendations, if provided, were given for each index.

CIE REVIEW RECOMMENDATIONS - DATA WORKSHOP***Conclusions and Recommendations***

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understanding of the parameter in question. It might also help if **a desk study was undertaken to examine whether the choice of discard survival has a significant bearing on the estimated status of the stock in relation to MSY reference points.** If the sensitivity of the assessments to this quantity is low, it might defuse some of the polarization over the chosen values.

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Assessment Workshop

- Investigate alternative approaches to age-length keys for estimating age from length
- Improve observer coverage, particularly during regulatory or gear changes in the fishery.
- Longer time series for surveys will always aid the assessment process. However, it is equally important to maintain the sampling methods and document them well for the most appropriate statistical analyses to be applied to the data.
- More time was necessary to complete the data vetting process for this many species, and in the future we strongly recommend that no more than probably two stocks be assessed simultaneously with the same number of participants.

Review Workshop

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Gulf of Mexico Blacknose Shark

Data Workshop

LIFE HISTORY WORKING GROUP

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COMMERCIAL STATISTICS WORKING GROUP

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- Improve observer coverage, particularly during regulatory or gear changes in the fishery.
- Longer time series for surveys will always aid the assessment process. However, it is equally important to maintain the sampling methods and document them well for the most appropriate statistical analyses to be applied to the data.
- More time was necessary to complete the data vetting process for this many species, and in the future we strongly recommend that no more than probably two stocks be assessed simultaneously with the same number of participants.

Review Workshop

The Assessment Team provided several research recommendations in the data workshop and AW reports, and these are endorsed by the Review Panel (RP) to the extent that they will improve the assessment. The RP considers research leading to an improved understanding of landings and removals, that improves consistency among indices, that reduces variability within the individual indices, and that leads to development or application of a model that more fully takes advantage of the length and age data including integration of the selectivity estimation into the assessment to be priorities.

With respect to further life history research, the RP considers the following to be priorities:

- Research on post-release survival by fishing sector and gear type should lead to improved landings and removals time series
- Research on fecundity and reproductive frequency should lead to an improved understanding of population productivity. As shown in assessment, status with respect to benchmarks is relatively robust to assumptions about overall productivity; however abundance and fishing mortality rate estimates are sensitive to this information. Research about natural mortality would also lead to a better understanding of productivity but traditionally has been difficult for most species.
- As noted throughout this report, the lack of age data was a limiting factor in this assessment and collection of sex-specific age and length data would aid the assessment. Regular collection of age data will help in the construction of improved age-length keys, in the interpretation of indices particularly in

cases where populations have spatially structured with respect to age, and significantly aid in fitting the selectivity within the models. Additionally, if the abundance indices are age-structured, population responses to management actions should be detectable earlier than if the indices only provide information on total abundance.

- Although information about stock structure is important, as noted under ToR 1, genetic studies may not necessarily be informative about structure. Tagging studies to determine stock structure need to take into account that populations may be discrete during reproduction, but otherwise mixed most of the time. Increased international collaboration (e.g. Mexico) could help ensure wider distribution and returns of tags.

With respect to the abundance indices, the RP recommends:

- Evaluation of the individual indices via power analyses to determine whether they are informative about abundance trends. The majority of indices used in these assessments exhibited greater inter-annual variability than would be expected given the life history of these species, and given this variability, may only be able to detect large changes in abundance which are not expected to occur rapidly. A power analysis would help to determine how much abundance would have to change in order for the change to be detected with the survey, and additionally, if the survey effort needs to be increased or re-distributed in order to be able to evaluate the effectiveness of rebuilding strategies given the relatively low population grow rates for these species.
- A small study on how to make the best use of the knowledge of the data workshop participants for developing index rankings.
- Ensuring that, to the extent possible, information about sex, length and age is collected for the reasons provided above.

With respect to the landings and removals, the RP recommends:

- Research that improves the understanding of historical landings, both in the modern and historical period and to support the assumptions about when stocks are at virgin biomass if this assumption is carried forward in future assessments. This is particularly important for GoM blacknose sharks given the difficulties reconciling the abundance indices, landings and life history information.
- As recommended by the AT, improved observer coverage particularly during periods of regulatory or gear changes (e.g. TEDs).
- Ensuring that, to the extent possible, information about sex, length and age is collected for each fishery in order that selectivity can be estimated in the model.

With respect to the assessment models, the RP recommends further model development using both simpler and more complex models taking the following into consideration:

- The RP noted that the models used in this assessment were reasonably suited to shark life history. However, other models (e.g. SS3) could also be adapted. If reproduction is modeled as a function of the number of mature females, uncertainty in the reproductive frequency, fecundity and pup-survival can

be integrated into a single parameter (the slope at the origin of the SR function), and information about these traits can be incorporated via priors on the parameter. The RP recommends consideration of this approach if information on reproduction remains uncertain.

- Estimating the fishery and survey selectivities within the assessment model.
- Development of a two sex model for more direct estimation of the spawning stock
- Fitting the model to either length or age data. In addition to being necessary in order to estimate selectivities, these data can be informative about changes in age-specific abundance.
- Exploration of models that do not require an assumption that the population is at virgin levels at some point in time.
- If external age-length keys are used in future assessments, development of a key based on a growth model to better assign proportions-at-age in each length class.
- Simulation tests (management strategy evaluation) can be used to test the performance of alternative assessment methods (including the catch-free model, ASPM, ASPIC, SS3, or stock specific models), recruitment parameterizations, harvest control rules, assessment frequency and data collection. Simulation studies may have a particular use in these assessments because of the particular biology of sharks and the data poor nature of these stocks.